ActInf Textbook Group ~ Cohort 1 ~ Meeting 3 (Appendix A + Appendix B +

Okay, it is May 19th, 2022, and it's week three of the textbook group, first cohort.

We're in week three, discussing Appendix A and B, and there are some notes in the sections

of the book, in the chapters.

There are also some ideas and questions that people have raised.

So, we'll go to the questions and then start with the most upvoted, so feel free to add

more upvotes if you want to discuss it, like even in this discussion.

And then, hopefully, if people are available to take notes in this section, that will help

add their thoughts in and also capture what the speakers are saying.

And then, we'll look at the question and then try to come to different answers and just

add more information that people can add more structure to later.

Okay, the first question says, Appendix A is described as the mathematical background.

So, maybe question one.

For the authors, or for you, what is the process of determining what is figure and ground for

the formalisms of active?

What other math concepts and formalisms are important for learning and applying active inference?

And then, three, what are some resources and approaches for learning math that help us learn what

is useful for active inference?

Okay.

Anyone can raise their hand, or we can just start to add some annotations here.

Like, what should be included in the primary regime of attention with the reading of a book,

either linearly, like some books are, or in a maybe moving around the sections?

So, what should be in the chapters?

What should be in the appendix?

What is not in the appendix?

What did people expect would be in the chapter, in the appendix, not covered?

Okay.

Yeah, Jessica, and then anyone else?

Hello.

Yeah, I was wondering about the multiplication of the matrices that we covered yesterday.

Which equations have the multiplication of the matrices?

Like, a couple of examples, just so that I can, like, play around with them?

Like, in the actual, like, active inference equations.

Does anyone know one?

This is referencing the way that the appendix A is starting with linear algebra,

and then introducing this operation of multiplying two matrices to get a product.

If someone can find, like, an equation from the, that we've already seen,
or some other equation while I'm typing up that question, that would be helpful.
Okay.
That sounds weird.

Can anyone
just describe what they thought the intention was of starting with linear algebra and using
8.1 as the first equation of the appendix or we'll return to just the more general questions
yeah um i mean linear algebra is kind of the most discrete um
i don't want to say fundamental but like uh practical and comprehensive way of kind of
working with a large space of data i guess um together and computing on it so it's kind of the
basis for the discrete uh parts
and maybe easier than the non-discrete parts good reason to start
i guess if the question was um where is linear algebra used in the active inference
and math um when you go to appendix b then you've got the equations of active inference
and these are all expressed in terms of large vector spaces of variables so probability distributions
and so for example on page 245 you've got this dot notation which is the expectation
of a value so that right that goes directly back to that first section of appendix a is what does that
mean in terms of how do you take an expectation of a large vector of things how do you express that
compactly thanks uh could you unpack we looked at the dot notation a little bit in uh yesterday
and they were mentioning how well i think it was actually one of the questions too let's just see if
someone asks this okay they say the dot operator in a3
the dot operator is equivalent to standard matrix multiplication where the first matrix has been
transposed so what is the relationship with expectation when we're talking like what are
where how does expectation this is for anyone how does expectation relate to linear algebra
the probability of a value so you're going to take all the possible values and you're going to multiply
them by the probability of that value and then divide by the sum of the you know well probability
of normals is to one so that will be the expectation so if you have a whole array or a vector of you know in
your distribution you can express that all in compact notation by saying with this simple way of saying
we're going to take every one of these terms and we're going to multiply it by the probability of that
term and the probability sum to one so then that will be your your average overall
awesome thanks for that answer
um
okay thanks for that awesome answer does anyone have any other thoughts on just this first question
and then we'll continue so what is figure and ground we're learning active inference in the chapters
that's why the chapters are there the appendix is there to somehow lightly supplement that like they
the first question and then we'll continue to do the analysis of the algorithm and the algorithm
and the algorithm is the first section a2 that is discussing a lot of important things like derivatives and probabilities
taylor series variational calculus and stochastic dynamics are going to come in in the coming chapters

but the linear algebra is is important for the upcoming chapters
okay so this question asked in a3
they say the dot operator is equivalent to standard matrix multiplication where the first matrix has been transposed which is flipped like it's the operation in google sheets or excel right click paste transpose
they say that in the case of column matrices
this is equivalent to the dot product
so the dot operator here is like a generalization of the vector dot operator the sum of the products of the corresponding entries of the two sequences of numbers
so the questions here were what
what is interpretation implication or use of the dot product of vectors
and then what is an interpretation implication or use of a more generalized dot product and there's looks like there's an awesome answer
what would be a situation where if anyone can think of one like
what would be a situation where if anyone can think of one like
the dot product would be applied like you know we're just we're trying to do this in this situation
this is the operation that we needed
we wanted to compute matrix a on something of something we used b dot c
um just continue eric's uh description from probability like you have expectation probability
and you wanted to kind of see the error or uh whatever the difference i guess uh between observation and and the expectation
then a dot product would give you
that answer or
i think it's a probability density of it
what are some ways to compute differences
like you can if there are scalars you can subtract like five minus three or something
then it was mentioned of cosine similarity does anyone think of or or imagine like another way that you could compute the difference between two different like kinds of
features of different dimensions
like another example is divergence
because that's like measuring the differences between these two distributions and there's different divergences the kl is one of them what are what other things are in this space that might be just tractable possible distance measures
well i mean these are these are the same dimensionality in all these cases but um
um another very popular one is the sum square difference some squared error
and then um you can have absolute different value difference so these are called different um norms
i think that's the right word yes there's also the genie index the jacquard index
um euclidean distances like sample to sample distances
um

so um could anyone talk about l0 l1 and l2 etc norms while i'm adding some links
the subgroups if i have the right numbers here l l2 norm would be the square root of the sum of squares
l1 norm would be the um average of absolute absolute value differences and l0 i don't know what that is
uh well i don't think it's a difference is it it's like a presence absence okay just so it's like a
that kind of encoding is used in a lot of different algorithms
and then the l2 norm is the sum of squares so the l2 norm is what is used like in a least squares
regression classical statistics t-test anova that's a lot driven by the l2 norm
and all of that is like in the genre of what the a3 notation is describing
and there but depending on what the b and the c are and what the a is etc there's just and the
dimensions of everything and other operations that are happening before or after it
later on they mentioned the quadratic which is the l2 norm so that's where you have the you know the
same term is multiplied by itself or if you have cross terms then you get the covariance
and that's going to be used later with the laplace approximation and other approximation techniques
any other comments on a3 or dot product or just like this like kind of linear algebra topic
linear algebra the basics
the trace and the determinant which are probably less important than the dot product but still come into
play the derivatives how things change with respect to each other
the other
possibly time possibly some other surface and then probabilities
so does anyone have any other comments or thoughts on that whole section like anything
that they read or like had uncertainty around up to equation a22
great that no one has uncertainty around any of the equations up to a22
yeah i mean obviously there is a lot to like learn and to understand and we're not going to
do it's just i don't know people who want to think about that like what are we learning in appendix a
why is appendix a there
that hopefully we can start to understand some of this in math learning group
the people who want to be really engaged with the math
questioning and process and then and also connecting it to computer science into the applications like has
been happening and then hopefully everyone can can benefit from this like finding the resources and stuff
because the relationships between the terms relationships amongst the terms
are driven by formalisms
like the way that preferences and expectations are framed in active inference
is related to equations not only a conceptual linkage
so we can see the difference between the terms of the language
okay
maybe here we can just write math learning group
and copy this to the resources
okay so what kinds of examples

do you think would be useful for learning the equations and concepts in appendix a

if the equations felt too general and not applying to anything in particular

what example or scenario or like would have made sense or you felt would have made that clearer

like we're running a business and first this person wants to do this

and then this person wants to do that or what example or what would somebody have expected to have found or wanted to have found

that would have made it clear

examples or other exercises

or a format this is kind of like a math lecture

it's kind of directional versus like what other things could happen where you would feel like you

were understanding and learning this and engaging with it yes thanks brock i was kind of reminded of a video that carl

i used to do a presentation um a presentation um where he like in real time demonstrates uh

uh and helps you

um work through a like a pattern of red like it's like three dots sort of that

are moving around red green and this pattern and so um understanding

like just a simple visual example

of how

active inference would work

that doesn't

require the

formalism to

just

the surface level kind of pattern

recognize. That would be

ideal, maybe, of

then bringing in the map

for this part and that part and

some scaffolding.

But

yeah, I'm not

sure the appendix, I'm not sure

it was meant to be bred in

I'm not sure there's a great way

to read one order

or another there.

Because you kind of need it, but

you kind of

yeah.
It's a lot too, right?
Yeah.
Great point and
like the
organizing team for the textbook group
which was open to everybody
who wanted to join the EDU
and the comms
weekly meetings.
So for future cohorts, people
could totally be involved with
planning it, communicating it,
doing anything for future cohorts
as a co-organizer or other role.
But we talked a lot about the reading order.
Would it make sense to read them
in
without
appendix or putting the appendix last
or somewhere else?
One was short
and hopefully the
one week of
regime of attention
on appendix A and B is like
we're just
skimming it
in a week.
we're not
getting a PhD
in math.
We're not generating
these equations
on a blank piece of paper.
We're just like
seeing the forms
that will be used

and then
some of the key
areas
in the order
that the authors
thought that those
background topics
were important.
So like for appendix A,
linear algebra
which is what we talked
the most about
because it's going to come into play
most
quickly
especially with these notions
of like expectation
and differences
which is going to come into play
with everything
that's going to happen.
Taylor series
variational calculus
and stochastic dynamics
we talked less about.
They happened in a later order
within the appendix
but hopefully
there will still be many questions
on them
because they're probably also
areas with a lot
to learn
and to clarify
and also appendix A
is like
not the final
resource

of it
diffusing
into us
like something
that was really important
that came up
in the math learning group
is
there's not
like a glossary
of variables
or table
of variables
so that's something
we can work on
with notation
is connecting
the variables
to
natural language
ontology terms
and then also
things like this
what do we even look up
or a symbol
or what does it mean
when there's a double arrow
there's a lot of things
that
like a lot of dots
that could be connected
if people
ask them
and then we can
probably get a response
like
what was the double arrow
in A3

what
does it mean anything
that they're offset
or what
is
what are the parts
of this shape
that are mattering
for the background
of learning
active inference
anyone can ask
like any question
if it's coming to mind
as an uncertainty
then writing it down
just anywhere
is really helpful
because then
on the first pass
through learning
or a
primary branch
of learning
we can just go
okay
here's what A1 shows
and then have that
in a way
that reduces uncertainty
more rapidly
than
how any of us
would just give
an ad hoc
explanation
Moritz wrote
one

thing that
I always
find useful
in thinking
about matrix
you know
linear algebra
matrices
is
to write out
pictures of the matrices
and their dimensions
because often
the notation
you've got these
i's and j's
and k's
and stuff
going around
but you don't know
how that maps out
into the actual
elements of the matrix
so
those illustrations
are always helpful
that are
you know
could be added
into this
this book
they're in a few
places but not that much
there you go
things like that
and
yeah
yeah

and also
like connecting
the programming
experience that people
might have
even in
no-code tools
like
excel google sheet
like
you have rows and columns
ones have numbers
and ones have letters
but they can be
other things
and then
anyone who's used
something like
python or r
or even just
doing some
statistical calculations
like if you put
two lists of numbers
into the t-test
that would be
vector versus
vector something
and then you would
address the fifth
element of the list
and then if it was
in excel
you would need to
address two numbers
to say where you were
in the matrix
and then the tensor

is just
any number
of more than
three dimensions
which is sometimes
like
it's like oh
but how spatially
are we going to
see it
but then
we're familiar
with working
with spreadsheets
and data sets
that are
larger than
two by two
or larger than
two dimensions
so
like you could
and that will
come into play
a lot
and the numbers
can be representing
probabilities
or be used
for probabilities
I also
Daniel
shared
in the chat
shared a link
to a python
notebook
that has

is a link
and it shows
kind of the
operations
on matrices
and kind of
you know
you can see
both the
mathematical
yeah
there you go
and so you can
see the actual
matrix
and you can
actually run
code to there
so
that particular
so
awesome
so I pasted
the link in
here
and then
we'll just
we'll
you can fill out
the rest
of the row
but then
someone could be
looking for a
guide on
matrix
or programming
and matrix

or something
before this
state
so then
each person
if they find it
and add it here
will have a lot
to share
and learn
okay
does anybody
want to
like if they
were the one
who asked it
this question
I think it's
probably referring
to long
equations
that probably
none of us
are ready to
give answers
to at this
point
unless somebody
wants to
contextualize
this question
okay
I'm hoping
a read
of chapter
four
will
elucidate

when the
time comes
just exploring
the formal
foundations
here
okay
okay
okay
does anyone
have a favorite
or recommended
textbook
that contains
the KL
divergence
and
Diersley
distribution
in the
index
this is for
the math
learning
group
okay
maybe we
could tag
some questions
too
and so
then we
can know
which ones
they can
look at
Wikipedia is
good

the KL
divergence
is approachable
the
Diersley
is tougher
the links
take you to
definitions that
enable you to
put the pieces
together
okay
any comments
on KL
and
Diersley
from someone
who is
familiar
otherwise
like it
sounds very
technical
and we
don't
need to
go into
it right
now
what is
up with
dividing by
sigma
in A31
also
detail
but

we'll just
see if
there's any
questions
that are
not
going to
be
specific
details
because
math
is
very
hard
to
understand
sometimes
on the
fly
and
there's
definitely
a space
for the
interactive
discussions
around
math
that are
unrecorded
which
are
important
however
this is a
little bit
of a

different
format
so
which is
how I'm
hopefully
interpreting
people's
activity
rather than
their
disengagement
because
they are
here
and hopefully
have at least
scanned
these
chapters
anyone can
raise their hand
at any time
or add
questions or
upload things
mean field
mode
equation 9
equation 16
probability
resource
statistical
question
appendix B
just to see if
there was any
also seem like

all details

so we have

23 minutes

what does

anybody want

to ask

now

or

is there

some

question

here

like the

most upvoted

one or

some other

one

that's far

less

challenging

that they

would like to

address for

the next

20 minutes

thank you

Thank you.

Thank you.

Thank you.

It is like crazy.

Like, like, I mean, even just in the appendix B, like the very is was at the very first thing, like, are they using Oh, for observations? Are they using Oh, for outcomes? Like I was totally unclear on that completely. So like, state inference Markov decision processes, like the states that influence outcomes, Oh,

Oh, right. Like literally, that's what it says on the top of page 244. So it says that the variable O is an outcome. Is an outcome an observation? Are those the same thing? Are they interchangeable? Like, I've always thought it was observations. I put that question in there.

It's already there. And then the question right before it, like this is a really tricky section, right? So there's two questions that are in Appendix B that are together. It says the likelihood of observations given a policy is not straightforward to compute.

This is because POMDP problem is structured so that policies pi influence trajectories indicated by tilde of states s that influence outcomes o without a direct influence of policies on outcomes.

The problem then involves a sum over trajectories of states to marginalize these out and find a marginal likelihood of observations given policies.

Like, what does that these even refer to? If you look in the questions in Appendix B, like they're there in the questions already listed.

So, I mean, I felt like it was very obscure and not super clear. And also like the notation gets crazy.

And I'm like really a stickler for like defining every single variable and symbol in an equation to understand the math so that I can actually read it in English.

So I was a little bit lost.

Totally agree. Thanks for sharing it.

Just with respect to that particular question.

I think that these refers to the trajectories of states.

And you can see that because the sum is over the tilde s.

So the tilde s is the trajectories of states.

So you sum over all the trajectories of states to try to figure out, well, what are we going to get if we apply a given policy?

That's my intuition about that.

And I suspect also that observations are outcomes.

They're sensory outcomes.

They're the outcome of a generative model or a generative process.

But they're the outcome that is observed.

But this is just showing how many dots we need to actually unpack and connect for unsighted, grammatically vague sentences.

Right.

Like, so thank you.

My uncertainty has been reduced substantially just by like figuring that out.

But like, I felt like that the whole time I was reading the appendix.

It's like, what is even going on here?

Yeah.

So in this.

Yes, sir.

The way I view it is like exercising.

You're not going to run a marathon the first time you set out.

But, you know, the more you stretch and warm up.

And that's kind of, I think, that's how I kind of view this going through the appendix is kind of refreshing on what you used to know about math, at least for me.

And so that's like stretching and doing a little bit of exercise the last week.

And then every time you go through it, every time you see the notation, you've slept on it some more, it becomes more and more familiar.

And so you can start to put together bigger, bigger chunks.

Some vaguely you can map to the notation and the math.

Other ones, it's still going to be dangling.

But that's okay because there's fewer dangling things that you're groping with at any given moment.

So I think it's great to go through these appendices and see it all through one pass.

And all right.

So I didn't run a marathon.

But that's okay.

I ran two miles.

That's pretty good for a start.

Thanks, Eric.

Another like metaphor is just like, oh, yes, please.

Jakob.

Yeah, I was just, I was just going to comment on what Blue said with the kind of some confusion about the notation.

Because I asked the question about the marginalizing of states, but then I was also confused about the O notation.

Because in equation B dot 1.

It seems that it's a trajectory over outcomes.

But then there's also an O without a tilde on it.

So does that mean that it's an outcome at a particular point in time?

But then there's also a bold O, which implies that it's kind of a vector.

But if it's a trajectory of outcomes, that also seems to be a vector.

Like a trajectory for me implies some kind of vector form because it's a sequence of outcomes given a policy.

But then what does that bold O mean if not trajectories over outcomes?

So just to kind of add to the confusion, I guess.

Thanks, Jakob.

I'm glad like we're grokking.

We're on the same page, you and I, for sure.

It would be awesome for description or any notes.

What is the reading or a reading in language?

The this of this conditioned on that is this over that of these two things.

The first thing is this.

The second one is that.

An example of that is in this situation.

Here's the script.

Here's the graphical abstract.

Here's the paper.

Here's the person you could ask.

Which one of these do you think is going to be interesting?

Okay, so Mike asked, and I'll copy this in.

For those who have read the entire book, did you find the appendices useful as you went along through the chapters?

Yeah, I also think that if the materials in these two appendices were to be integrated in the linear narrative of the whole book, it would be much more useful than separating out as, I mean, separate appendices, because as we see, I mean, that's my own experience. Well, reading the chapters independently and in isolation of the narrative doesn't give me a sense of their concrete applications and I don't know how to use them and how to use these for all of these equations and everything becomes much more fuzzy.

I hope this is not controversial that linear reading of the appendix as something that you look through is extremely confusing or extremely imprecise because there's many symbols introduced that might have been introduced earlier, like O is probably discussed earlier, but then it's briefly just sort of mentioned and then there's a lot of symbology that's not mentioned what it is.

So it's hard to read the appendix without prior knowledge, yet ostensibly Appendix A is the introduction refresher.

So is this the introduction offering?

The appendix to go some way towards remedying it, being the maths required to understand this not complicated basis.

The multidisciplinary basis means it is often difficult to find resources that bring together the necessary prerequisites.

The multidisciplinary basis.

They're going some way.

So this is like one kind of plank out there from their point of view from the chapters.

This is stuff and I also kind of agree like on what what ordering these are really interesting questions.

So then what is the next connector that picks up here with this artifact?

The next thing is, is that you can do a whole course on deep learning to apply most of the matrix operations and whole courses on like the math of them.

So they can't go into all this detail.

They can't spend one hour or like multiple coda pages on just what this means.

So there's has to be some kind of compromise, but then there's not going to be one specific perfect compromise.

With especially with like length and audience considered.

So I think like, I mean, I undertook like a very detailed linear reading of the book, because that's just kind of how I am of the appendixes, at least.

But I think for many people that like I mean, I've been there before, like where you're staring at math equations like this is gibberish.

But like, so just to read the text, you know, and like to read it and go through it, it's helpful to know that it's there.

So if nothing else, at least like you can be reading the book and then be like, oh, I remember kind of reading about, you know, Taylor series expansion in the appendix.

And so then you can just go back and just having access to it or the the refresh recall access, even though it is confusing.

I agree to try to a linear read of the appendix.

It's I think skimming over it or deeply reading it and then being able to refer back to it is useful.

Thanks, Blue.

So in next week.

We're going the next two weeks.

So the pace of one section per week might be like whatever it was for you, however much time you put in, etc.

The coming chapters are going to be very different because one didn't have any formalisms or figures, really, just the kind of overview figures.

But we're going to have two weeks for two.

So no need to rush it.

Read a couple pages and then just go back to the beginning and just restart the same pages like that's like the multiple coats of paint in a mural, like the low road to active inference.

That's where they're going to pick up in that high road, low road dialectic that we talked about last time.

Let's just see what figures, what equations might happen.

Okay, a lot of terms that hopefully are in the ontology already.

Like you can use the at symbol, hopefully to call most of them.

But if something's not there, we can add it to like supplemental or entailed.

Okay, so what figures and equations we might see?

Let's see.

Here's a box, 2.1, about probability.

Here's an equation that's the Bayesian kernel.

And it's going to be talking about this example of a frog and an apple and jumping or not jumping.

And that example is going to play a current role.

But then some likelihoods are shown in a specific example of like frogs and jumping or not and apples.

There's a work through example of exact Bayesian inference.

Here's a table on statistical distributions.

It would be really interesting to hear what support and surprise mean.

And why, what are these, are these all the distributions?

Are there other ones?

Or like why these ones?

Or why are they useful?

Where have they been used?

What does it even mean?

Here's where the KL divergence is introduced.

And then some more analysis from a surprise perspective, Bayesian surprise, with the KL divergence.

There's the box on expectations, which is also what we talked about a little bit today.

And that was like really interesting to connect it to the matrices.

A figure of the generative process and the generative model is the caption.

A figure, both perception and action minimize discrepancy between model and world.

A figure of the generative model is the result.

Variational free energy.

Variational free energy as an upper bound on negative log evidence.

The figure, but with equations in it.

Very common format.

Figure.

Complementary roles of perception and action in the minimization of variational free energy.

Big act-inf theme.

And like something, kind of a common fristinism.

Like perception and action being in the same game.

Or being in the same service of the same objective function.

Planning.

But no specific equations.

But probably citations.

Just introducing expected free energy.

Which Jakob and others can probably go into a lot more detail on.

Expected free energy.

About a future where the outcomes haven't happened.

Expected free energy.

Figure with equations.

The end of the low road.

Introducing the two key terms.

Variational free energy.

Expected free energy.

Summary.

What does anybody want to add?

Or what was something cool?

Whether they read it already or not.

Found it from anyone from

I don't know the distinction between the purpose of these boxes and I mean

what's the function of these boxes as compared to what we see in the appendices

because presumably these boxes cover the concepts that could be skipped over if one is familiar with

these mathematical concepts so do you have any idea about the reason behind this decision I mean

covering some basic concepts basic mathematical concepts in these

sidebar boxes and some others in the appendices

that's a great question a box could reflect like okay expectations got it or like okay

some product rule all right but it's kind of like you might want to read that or is this introductory

highlighting it this is the 101 on probability or is it saying this is a total skippable unit if you

want to learn about how the lizard does it here's where you look it can it can sometimes mean both

and like the the format of the book is also relatively austere though in a concise tone

like it's in black and white which especially in some later sections make some visualizations

where it's like it's hard to understand like where like what curves are doing what

it's a black and white image so what is there to see with the color I mean or but

it also might predispose towards more simple or visually accessible material so what is going to be

accessible and the order so those are all really important questions so we can just take notes on it

you know in the weeks that we're going to continue to do this because we only have limited live time how

about in the last like three minutes what does anybody think about the opening quotations or specifically

this quotation my thinking is first and last and always for the sake of my doing william james

you know i thought the guy was a philosopher so i guess you disabuse me of that one

yeah

these would be really nice areas to look into for people who like the history part um blue and i and

others are working on some different kinds of ways to reference papers and just up cool paper

sensation perception we have those terms we could just link the paper somewhere

so if anyone's like interested in that kind of architectures or that kind of philosophy question

those are the literature right now is small enough to know what has and hasn't been done

in a lot of areas that are philosophical and applied and technical

but just from like a first principles what does this seem to mean

uh that shared that other group that that we're kind of both a part of daniel um

um liminal dow thing that's not really a dow i don't know what it is but there's this debate for some
reason that's going on there about you know different ways of knowing and being and doing and a bunch of
stuff uh that is kind of a mix of philosophy and linguistic fallacies but um like if
if thinking is actually a physical process which i think you know i don't think anyone here is
going to argue against then it is something necessarily that is being done and
so however i guess small you want to draw that markov blanket up and then however large whether that's
your physical body actions or some extended thinking it's literally
what you're doing uh they're they're it's not just for the sake of your actions but it's
they are your actions like they're just others another set of your actions
all right let me i'll offer a different take on it yeah awesome william william james was a great visionary
and he foresaw mark's mark zuckerberg's meta metaverse 120 years ago or whatever and he said no
i don't want to live just inside my mind inside the metaverse i want to be out in the world interact
with real people and real things so he was a meta or facebook skeptic way before his time
okay thank you ali
yeah as brock mentioned uh i think it relates to an epistemology epistemological
uh i mean distinction between uh propositional knowledge and uh how-to knowledge and i think uh
this statement by uh this statement here uh tries to blur this distinction blurs the line between uh
propositional knowledge and know-how knowledge because in especially in analytical uh school of
philosophy uh there's always been a very um uh let's say um not heated debate but uh there's a long-standing
uh debate uh about uh about the distinction between these uh two kinds of knowledge and whether uh
they're in fact uh they can be uh distinguished from each other or not
thank you could you just unpack what are the two kinds of knowledge again and what are just they
referring to sure uh well about propositional knowledge uh well an example of propositional knowledge is
uh to know uh the exact um mechanisms of walking uh i mean uh which muscles contract and uh which uh i mean in what
angles uh the uh the uh i mean the whole thing about the uh biomechanics of walking the whole knowledge about the biomechanics of knowledge
uh can constitute this kind of propositional knowledge and uh it's totally different from knowing how to walk
uh i mean uh a three or four year old year old child has a know-how knowledge of walking but not necessarily
where he or she doesn't know anything about the biomechanics of walking so the biomechanics of walking
is the propositional knowledge uh actually knowing how to walk constitutes the know-how of walking
so when people say things like active inference is integrating perception cognition and action
maybe it is rethinking some of these long-held mental frameworks or distinguishing or operating differently
on action and perception like one fascinating recent example from me was and working with eric was
in the area of ant pheromone modeling without going into too many details people often modeled preference
as a function of the absolute amount of pheromone on trails because that's what the exponential like
decay is on that's what can be manipulated rather than the perceived intensity which might have a

different scaling relationship so it's like a dim room you can detect a small change bright room you can't detect a small change so that kind of psychophysics of perception gets ignored

and then you can see the difference in the way that you can see the cognitive

and the cognitive can be measured implicitly because of calls for measurability because the cognitive can't be measured directly even if you potentially had a electrical measurement or something like that happening

then these are awesome questions can you think without acting or is thinking in action

this active paper which was on a live stream so we can provide the link to

it models like perception and action in the sort of like kernel level just the autonomous sensing sentience bot level and then attention and metacognition are both related as actions

and then thinking in action and then thinking in action and then thinking in action

which is like why the paper is relating computational phenomenology with mental action

again just to have like a look at the kinds of models that can happen and then now does making any model fitting any data well enough

and saying well we modeled it as action so is thinking in action will that model that says yeah it's consistent with that will that ever constitute positive evidence for saying thinking is action

what's the problem or why even say that or what does it mean to say it

and then here's a funny meme

and here's the generative model in the generative um the partially observable markov decision process

here's carl friston

jessica and then anyone else

uh yes um i guess it's sort of like a beginning understanding of this is that

i tend to think that um

there's like a bias towards action in active inference uh i think like the first chapter is saying like

you know even if you want to like stance or like perceive you have to do some kind of action in order to

um gain the information and and this like you know quote is basically saying like okay even though

the whole field is trying to understand cognition it's like what we decide to do and the thinking that

we go through in order to determine our policy is to determine our actions and then it sort of like

feed into um the thinking itself so maybe that's why there's like the bias to action and

um because like our actions is what it's going to be allowing the thinking but um yeah like when

we update our models or like you know trying to come up with like the policies that we're going to be

doing like we have to be processing things so you know the thinking is in like as a service to action

and that's sort of like what i was thinking i don't know it's a little bit like you know with the

chicken and the egg but um you know i kind of tend to think of it it's like yes like we have to act

in order to understand but and like the understanding which will be coming from the thinking is what

determines our next action um yeah that's kind of what i was thinking

empath empath empath empath empath empath empath empath empath empath empath empath empath empath empath empath

apply in like a linguistic emergent term actions. And so we have to then pop outside of that and try
to consider with our internal mental processes, how to disentangle these concepts. But I see no
reason to think that, to think ironically, that internal mental states aren't also conscious
parameterizations of processes in the brain and our ability to grasp control of that to some extent.
And so it seems very strange to me to try to separate those linguistically from our current
vantage point, even though I understand why historically, working only with language, it
might have made sense. So those kind of quotes, like James, strike me as somewhat antiquated,
or just, you know, operating on dichotomies that don't seem to make sense, given that we've unpacked
these processes to the degree we have.
Thank you, Matthew. Very deep points. Like you mentioned, thinking as a verb implies action.
And then just to kind of complete that, thought is like the word or the noun form of this one.
So then what does that imply?
And then also like Jessica mentioned this.
It seems to imply to some extent that like we can only express processes through their discretization
of, you know, via symbols, right?
The idea that like there's this process, there's a flow, there's something that can't necessarily be
separated, obviously, from the rest of the flows around it. But to reference it, to point to it,
to describe it, communicate it to another agent, we have to nounize it, kind of like,
which is the opposite of verbalizing, so to speak.
Wow. Thank you, Matthew.
Thank you.
Again, these are, yes, Eric?
Sorry, I didn't have anything. I don't have the hand up.
sorry i didn't have anything i don't do it okay the hand was yeah so um carl fristen had a
influential 2019 uh paper long paper the free energy principle for a particular physics
so that was in 2019 so it's been several years since then and like particular
allegedly slash accidentally or intentionally is a pun so it can mean specific like a physics for
the specific systems that we're modeling or it could be specific like this is a specific approach
another interpretation is like it's for particulate entities when we define thing and the partition
which is going to co-instantiate like the generative model and the generative process and then the
markov blanket or the interface that's separating them that is separating the figure from ground
it's separating like the entity from the niche it's separating what from what what what can it represent
the separation of something from all possible separations some separations and then matthew you
mentioned like yeah verbalizing speech but it feels like we're speaking maths sometimes yet verbalization
is a process especially um dialogue yes matthew and yakub could i ask a question that brings in a little
bit of content from the other chapters of the book is that yeah sure i'm just kind of curious because when
you're talking about those um you're talking about the markov blankets um and integrating it into this idea
of how we draw boundaries i'm kind of curious uh is it fair to say that to the extent we've drawn our

boundaries and do see a reduction of free energy in this system it reinforces the idea that there's something to that boundary structure as an entity in the world that there's a reality to that is that a parent interpretation i think many people would have a lot to say and add this touches on like the question of reification of scientific models and the ability for an identified statistical model to be transcending itself and used in a realist way like ontologically real about true joints of the world and um i'm gonna go to live stream table and find several of the of those who explained their research on this topic but even even sort of trying to claim it's real like is it just
does that ability like once we've drawn a boundary um using these markov blankets uh and we do see that it seems to be let's say using information to um uh self-evidence do we think that that's i mean it seems like the the the the this you know this paradigm of thought is predicated on the idea that um to the extent that that occurs and reduces free energy it should attract or further attract attention at the very least uh for examination or investigation yes so great questions yakub do you have something to add on that or or is it a slightly different area i guess i guess it's kind of related um to the to the mark of blanket discussion um i know this may this may be completely wrong but it seems to me like with um um that this kind of discretization of the mark of blankets forces us to a specific kind of discrete thinking but i feel like that just this model of a of a markov blanket i think of it as a kind of classical model but i presume that the reality is more something like a schrodinger markov blanket where it's not we can't really draw boundaries between thinking and observing and generative process uh in the same way that we can think of um say electrons as balls bouncing off each other but they're not really we can think of them as classical particles and that will help us um to an extent but then we can also observe um the process of thinking so thinking is also kind of entangled with the generative process so there's so even though the markov blanket is a useful formalism i think that there will be that there are definitely certain scenarios that um in which it's more like a probabilistic well it's already probabilistic but in the sense that the entity that's performing inference also in its own inference loop performs inference on itself and therefore thinking is at the same time an action but at the same time it's part of the generative process
i think i think i think i think i think i think i think i think i think i think i think i think
i think i think i think i think i think i think i think i think i think i think i think i think i think
these bed stretch it out so that this perception and then active as a filter is was reminding me of what matthew was saying like going out and discovering divergences from expectation of there not being metabolic activity on a planet and then there is some statistical deviation and so like in the classical statistical framework that might be like it's a one sigma difference in the bayesian framework well there's a base factor of two and here for this evidence and there's other ways to like talk about detection of novel entities that might be part of like a reification process of the extended cognition of the modelers
and then like one philosophical angle on that and then also anyone who who um wants to read then ali is um this book by helen longino which talks about methodological pluralism and about how there can be like disciplinary rigor and the ways in which often unstated like social

priors can be the substance of what becomes understood to be science as like a complex phenomena

um ali and then brock well adding to what yakov said um i i think the the concept of markov blanket

emanates from the uh plutonian uh way of thinking uh specifically uh hylomorphism uh school of thought uh as

uh as opposed to hylozoism uh well the distinction between hylomorphism and hylomorphism is that uh well in

hylomorphism everything is disconnected every concept is disconnected and uh with every other concept and so

there's a discontinuity uh there's a discontinuity uh metaphysical discontinuity uh between the concepts uh but um

in hylozoism uh the philosophers uh advocating this uh school of thought uh well they claim that

hylomorphism fails at uh answering uh the question of what it is adequately and so and uh they even

rephrase the the question and they claim that the right question to ask is not what it is it's what it

can do so that's basically the distinction between hylomorphism and hylozoism but i think uh the markov blanket

is a way of um formalizing this hylomorphic way of thinking at least that's my opinion

thanks a lot for that ali eric

sorry i'm not hearing eric can others hear eric

maybe i need to reload

come here okay i'm gonna reload

i don't think i can hear

um okay i'm gonna okay wait eric try again okay how about now yeah now it's now i can see you in

here again okay how about everyone else who's the audience okay wow disrupting the um on the high

low field it's happened before um i would i just throw my two cents about um my interpretation of

a markov blanket it's about trying to make um to perform simplifications that make computation

tractable um if you have everything interacting with everything else then we can't do computing

on that so we um we apply uh compartmentalization and build objects and interfaces for how the

objects interact with one another um and those become tractable that's what um you know what a

bay's net does is it says what are independent and what are dependent you know variables on each other

and um that works well when those abstractions the objects and relations the nouns or verbs

are a good fit to how the world actually operates so that is a driver for learning or building

representations that will use these markov blankets in and put the compartments where they actually

are faithful to the way to the compartmentalization that we can abstract over the way the world parts

in the world operate so things that are distant that separate uh that interact only weekly we try to

maybe factor those out pretend they don't interact at all or use some other intermediate variables to

to represent the interaction in order to simplify things so we can do computation so that's that's how

i think markov blankets and we'll see if the math tells us that later on when we get to it

awesome brock and then nally

okay i'm like not hearing brock unfortunately do you yes we all hear brock brock is reloading i think

now again okay yeah gather does this sometimes but i don't know if it's if it's because we have 15 people
yeah but it's yeah like matthew said i don't see i don't hear him it just like there's different drop
offs um i can i can hear you but i still think daniel can't hear you daniel you can't go for it okay
continue interesting can you hear me yeah oh okay um yeah no this is this is just this ongoing thing that i
don't know we just keep keeps coming up about yeah well okay maybe they exist maybe they don't markov
blankets really but like how would they form in any um like how would they form and evolve and collapse and
presumably there is some point in in that process this like evolution part where uh the markov blanket is
is extremely poorly defined uh like relative to the system right and that's basically these sort of
underexplored areas it was was also in relation to matthew's question also about um the free energy
minimization thing where there's systems like non-equilibrium systems basically where um
um still something interesting worth you know worthy of study is happening but perhaps is not um
free energy minimizing um at that in that state
thanks for sharing it just a few thoughts like um just because so let's just say some sort of
mesoscale or global as that's called but that doesn't mean about the whole world but some sort of
global free energy minimization in the joint model is achieved like in a conversation that isn't the
same thing as the local disconnected free energy minimization otherwise fitting high parameter models
optimally would be simply reducing to fitting single dimensional models alone if we could just do the
linear optimization on the standalone variables then why would we ever need the larger dimensional methods
and then um like think of this textbook maybe with where things could be in the coming years
the one the the the markov membrane is like the linear algebra and then jacob mentioned that this is
classical model it is kind of classical in a sense in the the timeless classic sense so the one layer markov
blanket when thinking about like the rise and fall of civilization is not the end all that's like i think
maybe analogous to a linear regression y equals mx plus b and then you have like the high all these
100 years of development on the linear regression model and all these techniques and applications and
pipelines so the markov blanket one layer potentially over interpreting it as already presenting with
strengths or weaknesses in certain situations when not being just like empirically demonstrated in the
general case we'll see what could be said and when matthew
yeah um along those lines i was kind of curious i've seen that there's been a decent amount of work
on the sort of hierarchical or fractal composition of these models um i'm curious if that implies that
there's a fractality to the boundary by default or if there are any or like is there any specific work on
sort of hierarchical composition of the boundaries of that so-called membrane as you said
with better annotation we could have better answers because there's many papers that we've discussed in
a guest stream or a paper and there's also like many other papers obviously that that we don't have that
kind of annotations of um because it's quite common to see nested models in the context of nesting of
cognitive processes like in that sandved smith paper which was i think number 25 that was about cognition
as mental uh action so that was nested and the nesting was interpreted as cognitive actions and
counterfactuals basically and then sometimes nesting is used to refer to actual like well the state is
inside of the country and then the region is inside of the state like it's a it's implied that the map

is mapping onto the territory maybe even spatially the cell is inside of the tissue and that's like answering schrodinger's question everything happening with the planetary scale analyses cellular level um and then this is hopefully what is uh spoken to with like the composability of active inference and sometimes that's frame in terms of like the the lateral composability of like how you could have three ants interacting or 300 ants and then the the nested composability you could have the fifth and the seventh and all those layers with computational trade-offs maybe with no extra information to be gleaned and then learning the structure of the generative model or just the structure of the partition more broadly and like what the variables are and everything that's the structure learning challenge and cognition as structure learning hashtag synergetics geometry of thought is what fristen has and others have raised as like a total open area because there's the parameter fine tuning once you have the base graph so then you're you're now inside of the ability to like to reify or not that model you're just doing parameter optimization you know we used two factors in this linear regression and then we optimized it with the l2 norm two is the best number of factors but the the it couldn't be set at that scale you could say in a two parameter model this is the best parameters with this norm but then you couldn't pull back another level and so that's like meta scientific and meta bayesian analyses which they're going to talk about like later okay yeah i guess i'm also kind of wondering if there are any uh examples that come to mind of something like a system that is structured such as the clearest or simplest example that comes to mind is like let's say that you have a nation and a state within that nation and there is a port and there is an overlap of the sort of that port interface to an outside structure that is shared by both the national and overlapping national and state interests and decision processes um and so i'm kind of curious if there are examples of demonstrating that kind of composability of generative models um or if that's something that's still um just new and open those are awesome questions if people know any um some areas to investigate like in the phenomena that you're you know overlapping interests informally there would be like shared regimes of attention there's the question of synchrony and that doesn't mean like um synchronization identically but generalized synchrony then there's like coordination of affordances we're not going to do this because of that so that's like um in an area that that jacob and others have been working on is just describing that situation as an affordance on an affordance or like e sub e like it it's not in a textbook but you know if people are staying this long then this is just kind of some ways that some people are thinking about it and these are the areas where people can also do research and learning with us but like affordances on affordances the safety could be on or off so then there's like an affordance that modifies another affordance and could those have compositionality just like some of these other um generalization affordances for generalization basically what are those and then isn't it like discovering what those are just really i think uh it's kind of also related to that question of this formation evolution collapse of markov blank is this when when does the observation actions of one system map to the to the preferences of another and vice versa how does that come in to being existence how does that start to happen and how does that unhappen i don't know awesome question thank you and and another like angle on that let me just check it um and then ali is

like this is a graph that we're going to look at a ton and we will interpret what all of it means

what is the o a s b pi g etc but this is a bayesian graph like eric mentioned and the edges are like

dependencies not causal influences in the world but apparent potentially different interpretations is this

the only skeleton no this is the y equals mx plus b of skeletons and there's the the linear regression in

the first equation in the textbook of the stats textbook and then there's all these accessory tests then

there's like creation and destruction as applied maybe there can just be a loop that's not active inference

just checking every day if something's within an arbitrary threshold so there could be some

liminal or gray area like an interface i think steven sellette mentioned it as like a nail bed or

something like an interface between more particulate and then less particulate more more field versus the

particle ali uh well as a little side note um uh there's a popular science book coming out i think in june uh

namely the romance of reality by bobby azarian uh which uh touches on uh the uh uh the emergence of markup blankets

uh the dynamics of uh emergence of markup blankets and it uh attacks um attacks it from many different angles

evolutionary angle uh i mean uh from a cosmic even uh perspective and uh it even goes for uh goes uh uh as far as uh

uh well considering the whole cosmos the whole universe uh as uh let's say a kind of meta meta markup blankets uh so to

speak and uh well uh i think uh that could be an interesting thought uh put forward by bobby azarian in this

all right thanks daniel thanks um in fact he gave part one of what was uh intended and may still be like a multiple part

discussion but this was him giving that presentation

so we were in contact is definitely you know it's an interesting view and it's going to take all kinds of

of research and education and communication jessica

yes i mean i guess um

when i like first like join the lab like some of the sort of like um ideas that helped me a little bit like

to want like to start even like conceptualizing this uh i mean in addition to the filter that you mentioned

before daniel and like how like it could be like more like porous or a lot like or more hard and allow

things in and out but it was also um about like it could be like um just grouping of relationships

and interaction so like the closer like a relationship like different things are um maybe that's more like

on biological systems i don't know like if they're closely related and or they have like more

interactions even though they might be different but like grouping it in those sense uh and maybe that relates to what eric said

on from the calculation apart

and but that was sort of like you know some of the things like where i started connecting on mark of blankets to

uh sort of like okay if

you know items or like different objects and things like they're closely related um you can start like putting on like a blanket around um and maybe go so like with idea of wrapping things um

from some of the definitions but yeah those are kind of like the kind of visual ideas that i

you know started like helping me a little bit uh and i still like don't understand properly but

those are like the things i was like okay so relationships filters um like

like kind of like connecting interactions and between things

awesome thanks a lot for that like interacting entities so you could have the edges representing some

interaction like when people are fitting a linear model interacting variables so there's this whole

discussion topic of whether the interactions are like the two ants bumping into each other whatever

that means in the quote real world and then there's like the statistical interaction so then defining

certain variables in statistical models so leaving that debate behind us and just talking about the

statistical models that we have bayesian graphs some variables have edges between them that could be either

like designed to there be there or not or you could do some sort of like thresholding approach and

explore that your thresholding parameter was acceptable but like the more of a loose interaction you allow

the the more challenge there is to fit that statistical model and you may not have enough data to fit like

the 10 variable by 10 variable with all the interaction terms which is like why we're doing linear modeling in

a health population example they would like do model selection on what their statistical power is with that data set to resolve certain kinds of effects and correlations and like non-sphericities

and that is addressed a lot in the spm textbook and for instance earlier like pre-act-inf work but then you mentioned like

how to like how to like be engaged with that process of like the the wrapping or the seeing the clusters and

then knowing like where's there going to be like maybe generative models arising that are consisting of

other generative models or other generative processes and then like some things interact more with others

statistically in the model as by design or just as an outcome of whatever so there's sparse connectivity

amongst the variables they're not like all by all connected and then that sparse connectivity

simplifies a lot of things that's also related to like a lasso regression and also to the below l2 norms

and then a sparse model can be factorized and that is what allows for the variational bayesian inference

which is like doing bayesian model fitting on a factorized because it's sparsely connected graph

and so a lot of these discussions are quite downstream of a lot of the philosophy of map and territory

but it's still super important conversation however within the model inference with a model inference using a model

a lot of these questions are very technical and downstream of importance qualitative things that are also important to keep in mind

but of a different type

jessica or anyone else

yeah wow very interesting

here's one other question i guess we can discuss like as the as it currently stands there's a one hour meeting

and then at the end of the hour the very next hour begins the dot tools regular organizational unit meeting

so is there i mean it is the people who are here now but hopefully others would be listening to it if
they're not able to make this time but like
what will help people with the synchronous and asynchronous make the most of the next few months
do people appreciate having a longer discussion
discussion

do people think that the main versus the math group
like is there another subgroup like a philosophy discussion that people want to like kind of
really do this
because we're not like
we don't we're experimenting with open-endedness on our first cohort and in these early phases of the book
we're going to be able to do that in the early phases of the book and people can probably imagine various of like the things we want to balance
like respecting everyone's time
and different backgrounds respecting their preferences for
how much they want to learn about different topics
being realistic about how much asynchronous and synchronous
direct and peripheral
time makes sense
but also being realistic like
like
there isn't a two minute video or a process
so how do people think about like that
those who have stayed
this far
you

you
you

!
Mike and then anyone else
can you hear me?
yes

okay

um

yeah I think that

as noted earlier in the discussion there are just so many themes that run through this

that there are opportunities to

pull on any of a number of threads in the course of discussion

and so

uh

today's discussion was interesting and I stayed

for this part of this the discussion

simply because of the interesting threads that were being pulled

um

certainly a contrast with the first part of the discussion around math and I think a lot of uncertainty

amongst the attendees about

how to engage with the mathematical aspects

and so

maybe to try and put a point on it

um

from my perspective

learning about

what is active inference and how can active inference be applied in real world situations

is a motivator for me

participating

um

obviously there are also sort of philosophical and maybe more scientific discussions

that can take place

um

and

so

any

any meeting could touch on

any of those aspects

as well as

some of the asynchronous interaction

could pick those up as well

so

thanks a lot

okay anyone can raise their hand

here's just a few other
options like if somebody
hears something that's interesting to them
we can
in the discord
make a channel
that is
relate or people can participate in the regular channels
just like questions like or
and that's going to potentially be
seen and interacted with by people more broadly
like what if we posted questions that we're having
we're in the cohort one of the texts group
and we you know had this question
do you know
that's one option
another
um
and this is like to mike's
um
expression that applied active inference is a
motivator
epistemic and pragmatic value
expected epistemic value
we expect to learn a lot by like staying
in these
sometimes
challenging or oblique or whatever discussions
but then there's also expected pragmatic value
with applying active inference
learning and applying active inference
so then
maybe that is a group we can partition
like
an applied active inference group
so then
we can
be clear about what the focal artifact is

because also we want to like

not

all groups at all times

will be able to have

nor is it useful to have open-ended

discussions of any discursion length

so like knowing how far and in what ways

and how many minutes of people's linear time

um

ali

yes

the edu meetings

just one note on the organizational unit meetings

the organization units in the actinfo lab like edu

comms and tools for education communication and tools

are like directory

one hour

lab meetings

or group meetings

because we're not doing

the education work necessarily in that one hour

but some groups sometimes it's possible to do some stuff

but it's like

increasingly moving towards

sharing updates

from people who want to commit in asynchronous work

or smaller groups that want to commit to doing something like that

so then

educational related projects that's their opportunity to

ask for help share updates

and so on

but

earlier on there was more topical material

but then

through particularization

and operations

and other approaches

it becomes more like pragmatic

and less discovery
and less mixed media
role specification
all these processes
so
people sticking around is sort of the
and being engaged and like seeing an affordance
and then just making that contribution
wanting to be a facilitator for a certain project
or wanting to contribute actively
or just even connect with another participant
like you can email them
if they've provided their information
just some random things
but like
we want to have the applied angle
the philosophical loop
and how to even partition that discussion
like we couldn't just overgo every math
every philosophy
every applied question
and have
the judge and the jury
and all this apparatus
so
how can we scaffold that conversation around
applied active inference
for the people who are like super excited and motivated
mike
yeah i just want to add
there's an interesting duality related to what you just said
in that we can't go into fine detail in all of the content
and at the same time i've found this group
to be remarkable at unpacking things
and sort of
really getting into
what do we mean when we say that
types of discussions

and uh
what does this term mean
not taking things for
for granted
as going through the text
and
so
there's a balance to be struck
in taking that approach
of
asking what do we mean when we're talking about this
and putting nouns on things
and relating it to language
and not going too far over into getting into fine detail about things
and then i'll leave us your question
and then i'll leave us your question
yeah this has been
this firstly a great session
really great conversation
i really enjoy it
this is new
much of this
most of this is new material for me
and so
i'm really enjoying the breadth of the conversation
and i'm cognizant of this
aspect you're digging into is
you know how much do we
sort of separate out to pieces
and go deep dive in different places
versus more of a you know touching on multiple threads and their interactions
for me while i understand there is a balance to be struck there
i'm really enjoying the challenge of relating as an example the math to the philosophy
to all these different threads together because they are linked right
and and so i do understand that you can't
some of those
some of those areas
need a separate group that you can drill down into

for me personally and so this may be not the same as other people in the group
i'm really enjoying understanding the connections
the philosophy
how these different thought processes came to be through history
and get that depth of understanding
the philosophy of the world
thank you um lyle
and for those who want to like listen
or view
these live streams have many many themes
so check if there's papers that you're interested in here
the live streams is about papers
so there's 46 papers
guest streams are not driven by a paper
sometimes the person's sharing a paper
but these are like presentations
hearing from different perspectives
and if anybody wants to
help organize these
that's what we do in comms
if they want to facilitate
and participate in these discussions
if they want to like contact authors
recommend papers
these are all distributed tasks that
are leverage points for people who care
to just do a ton of amazing things
like if somebody's interested to connect it to a given
community and make an artifact or live stream
co-organize that like
we can catalyze it at the lab scale for individuals who
know about the affordance but then want to take that affordance
to just have really leveraged impacts in the active ecosystem
so all these questions and then jessica
you asked about the timetable for the edu meetings
like what specific times are they occurring at
the edu meetings
and then they're on mondays at 13 and 23 utc

there's two edu meetings to reflect like education being the primary mission
of actinflab and then that spreads them out by time zones
but if somebody really wants to contribute to an area
the organizational unit meeting is not the rate limiting step
anyone who has attention to contribute will be able to find a regime of attention that is
like connected to a task that's meaningful
the rate limiting step is not people's availability for a one hour meeting
it's however much time people want to contribute whatever practices and things
will find something that works for people who want to be engaged
so don't take these meeting times as being like when we're doing it when we're deciding even
anybody who wants to can email or contact just the lab email address
and be started on figuring that out um jessica
um yes this was related to the question about applied
um projects um i was thinking that maybe one thing that we could do is um on the project
and ideas on second year to have a table and where people can like you know briefly share what they
want to do or like you know just say like okay i'm interested in machine learning and apply you know
like active inference and maybe this specific topic who will be interested in discussing this
so exploring what to do and and then like adding the names of the people who would be interested in
that and so the person who started the you know seeded the project then they can be you know can start
like maybe contacting those people and see like you know what time they can meet and start like
creating their own subgroups um so maybe that's something that we could do like to facilitate that
um in a simple way um yeah so just start like saying like this is what i would like to do apply active
inference while also studying the course and you know we like to connect with people here who might share
the same interests on and you know see what kind of feedback on that person can get thanks and just
connecting and building trust and having like a buddy system or small groups or just people who are
on the relatively long path to apply active inference on teams we don't have the speed dating hot swap
active application protocol but connecting with people and then
however it's authentic in that relationship seeing what you're interested in in three months
will be in a pretty different situation but we still won't have even gone through
the second half of the book on the application of the textbook so there's a there's a lot of time for us to
to to develop ideas and to connect with each other
so thanks for sharing that a lot jessica and and i tagged you so that we can create a table
with the right way or or do it with however is the right way because it it helps enable connections for
people who want to connect around applying and also it helps us remember these are the specific
reference points in the text that are like our attractor regime of attention for this textbook group
like the textbook group isn't all of active flap if people want to apply active inference it doesn't have
to be from the ground up but it totally could be but there's many ways to apply that are just in
different codas that people get can get involved in immediately so if somebody feels like doing things

i hope they feel like they have the agency to do that

okay any final thoughts on this interesting

semi-pattern breaking

and also um yes one final thought for me is uh we would have had dot tools organizational

unit meeting at this time which is why near the end like i asked if you want to do different scheduling

because um in general it's probably not good practice or ways of working to like go over time

to respect everyone's time and all of these types of things but also in the future we're in gather so

people who want to keep discussing the textbook could go there into a different room people who

want to do tools can go into another place so we need to figure out how to

do that through the people who want to be there like these people and the feedback that everybody has

in the ways they want to co-create it tim and then anyone else who has like last thoughts

hey yeah i was just going to say this is generally way more interesting than the tools meeting generally is so

this cross-pollination shows great potential already i would say um

yeah sounds good i just wanted to add to that conversation though about uh someone even mentioned plato and socrates and all that

uh uh and that whole sort of recursive involution of the of the sort of uh that sort of uh

reification thing you were talking about and and sort of the the the thought processes maybe being

almost like a recapitulation of the of the of the of the structure learning and that and the creptasian

factor graphs and all that kind of thing you were talking about there um plato had like a concept of

knowledge which which he referred to as recollection and and this idea way way back where all all learning

and knowledge is actually an act of recollecting what we already know and i just wanted to point out there's

a maybe just more of a interesting uh illusion or connection i guess between those two where there

seems like a old new what's what's old is new again or what happened covering thanks tim

anyone else who hasn't spoken or who would like to add anything

anyone else who will be talking about it okay anyone else who will be talking about it okay

Okay.